CogniSplit: LLM-Mediated Instruction Distribution for Effective Collaboration

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Collaborative tasks often place significant cognitive demands on group members, particularly when instructions are complex. Prior work has shown that distributing instructions can reduce cognitive overload and improve team performance, and separate research demonstrates that large language models (LLMs) are effective at instruction generation and summarization; however, these lines of work have not been combined to distribute task instructions using LLMs in group settings. We present CogniSplit, an LLM-mediated instruction distribution system that decomposes complex collaborative tasks into distributed instruction sets assigned to each team member. We evaluated CogniSplit in a controlled study with 36 participants organized into 12 groups, with participants completing collaborative LEGO assembly tasks. Results show that CogniSplit reduced initial assembly time by 77.29% and reduced total completion time by 48.61%, indicating faster coordination and lower cognitive burden. Our findings position CogniSplit as a proof-of-concept toward deployable instruction distribution systems for collaborative work.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; **Collaborative and social computing**; • **Information systems** → *Collaborative and social computing systems and tools.*

Additional Key Words and Phrases: Distributed Cognition, AI Augmentation, Teamwork

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1 Introduction

Teams often face complex collaborative tasks such as cooking and furniture assembly. Classical instructional approaches for these problems provide a single comprehensive instruction manual that each team member reads in its entirety, describing all steps necessary to complete the task. These approaches require individual group members to process

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information that is not relevant to the specific sub-tasks they may need to perform, incurring an unnecessarily large cognitive load on members as they attempt to distribute tasks among each other [9].

In contrast, pre-distributed instructions deliver only relevant information to each member. This approach emphasizes optimizing distributed cognition: no single member possesses complete knowledge, yet the group collectively holds all necessary information [8, 12]. While distributing roles improves efficiency, the act of creating and assigning these roles creates a coordination bottleneck itself [8, 12].

To address this challenge, we developed CogniSplit, a mobile application that leverages Large Language Models (LLMs) to automatically decompose comprehensive instruction manuals into distributed instructions for each group member. We evaluated CogniSplit through a user study in which participants assembled LEGO sets under different conditions ("Distributed" vs. "Complete" instructions). Our results demonstrate that CogniSplit's distributed instructions reduce perceived cognitive load and coordination effort compared to complete instructions, highlighting the potential of LLM-mediated instruction distribution to support collaborative work during initial coordination and task execution.

However, our findings also surface important limitations. Particularly, our system fails when participants are unable to or choose not to complete their assigned instructions on their own. For instance, attempting to "un-distribute" the distributed instructions in order to gain a better shared understanding of the task amplifies coordination overhead and cognitive burden. Additionally, efficiency depends on the structure and granularity of the distributed instructions, as poorly scoped task splits can create confusion or idle time. These limitations motivate future work on better scaffolding and interface constraints that support role adherence, as well as adaptive role assignment and personalization.

2 Related Work

Cognition Theory and Costs. Cognitive Load Theory establishes that the mental effort an individual can expend during problem-solving is finite [18]. To operate beyond these individual limits, Hutchins et al. [8] proposed the framework of distributed cognition, famously utilizing the analogy of a ship's crew: while each sailor performs a limited, local task (e.g., mapping a landmark or raising a sail), their combined, distributed efforts enable the complex operation of the entire vessel. This framework suggests that cognitive burden can be successfully "off-loaded" onto other people, artifacts, and the environment [7, 12, 17]. However, this offloading is not automatic. Payne et al. [15] highlighted a critical constraint: users are adaptive decision-makers who only choose to offload cognitive tasks to the environment when the interaction cost of retrieving external information is lower than the effort of internal processing. If accessing information is difficult or requires substantial effort, users may revert to relying on their limited, error-prone internal memory. Therefore, to encourage effective offloading and reduce error, systems must be designed to minimize these costs, optimizing the value of information when needed.

Technological Solutions. To minimize this interaction cost, we look to the framework of Distributed User Interfaces (DUIs), which allows system designers to partition information across distinct dimensions (e.g., space and time) [5]. By utilizing *spatially* distributed instructions, systems can reduce the extraneous cognitive load associated with navigating between stacked or hidden information layers [9]. Furthermore, physically separating information across multiple displays has been shown to improve individual task efficiency by allowing team members to work in parallel [19].

LLMs as Orchestrators. While distributing tasks reduces individual load, doing so indiscriminately poses a risk. In particular, fragmenting the core idea of a task can prevent teammates from maintaining a shared understanding [3]. Therefore, effective distribution requires intelligent synthesis rather than simple fragmentation. Recent advances suggest LLMs are uniquely positioned to handle this orchestration. LLMs have demonstrated the ability to outperform humans in summarization tasks [16] and autonomous task coordination [13]. Although challenges remain regarding

the reliability of instruction synthesis [11] and managing task transitions [4], LLMs have successfully demonstrated the ability to facilitate information sharing [1] and provide frameworks for interpreting group dynamics to support task completion [10]. By leveraging these capabilities, our work utilizes LLMs to synthesize comprehensive instruction manuals into distinct subsets, organizing them across space and time to reduce the cognitive load of the team.

3 System Implementation

CogniSplit (pipeline and app interface in Appendix F) is a mobile application built using the Ionic Framework and JavaScript. The system operates by ingesting centralized textual instruction manuals, often found in real-world collaborative tasks like cooking and furniture assembly. By processing the complete manual, the system provides the LLM with global task context, enabling it to reason about structure, dependencies, and parallelization opportunities. After users input the manual via text or an image and specify the team size, CogniSplit prompts the LLM to generate distributed instructions for each member, containing role-specific tasks, explicit coordination checkpoints for interdependence, and a shared description of the overall goal to maintain common ground (system prompt in Appendix D).

4 Evaluation

We recruited 36 participants (aged 18–23) split into 12 groups of three, with generally moderate self-reported LEGO experience (75.00% "moderate", 22.22% "high", and 2.78% "low"). We utilized a within-subjects crossover design with two conditions: *Complete Instructions* (control: all members receive full text) and *Distributed Instructions* (experimental: instructions segmented uniquely across members). To control for order effects, instruction type and task assignment were counterbalanced using a balanced Latin Square.

Procedure. Team members were seated side-by-side at a table, with disassembled and randomly shuffled LEGO pieces in front of them. Each team was given 5 minutes to meet but was instructed not to discuss strategy, and then completed two trials using the CogniSplit app, incentivized by an asynchronous competition that offered cash prizes ($150 for first place and $120 for second) for speed and accuracy. Distributed instructions were pre-generated through CogniSplit prior to the study and standardized across all groups to maintain experimental consistency, and were manually validated for accuracy. To prevent practice effects (learning the build structure), we utilized two distinct but comparable LEGO sets ("Red Jet" and "Red Boat"), with textual instructions provided by Bricks for the Blind [2]; these sets were specifically selected for similarity in piece counts and assembly logic. After each trial, participants evaluated perceived workload using the NASA Task Load Index [6], and after the final trial, completed a comparative survey and a semi-structured interview to discuss coordination strategies.

4.1 Results

Task Completion Metrics. We measured three performance metrics for each group. *Average first assembly time* captures the mean elapsed time from task start until each participant first connected two LEGO pieces, serving as a proxy for initial coordination effort. *Error count* records the number of misplaced or unused pieces, normalized to errors per 35 pieces to enable comparison across the two similar tasks. Finally, *adjusted completion time* combines speed and accuracy:

$$T_{\text{adj}} = \frac{T_{\text{completion}} + E \times 42.86}{60}, \tag{1}$$

where $T_{\text{completion}}$ is the duration from task start until a team member announced completion (in seconds), and E is the normalized error count. T_{adj} is measured in minutes. Additional methodological details are provided in Appendix E.

Figure 1 summarizes these measures for the complete and distributed instruction conditions. Each sub-figure reports results aggregated across all 12 groups (solid bars). Each sub-figure additionally reports results aggregated across the 8 groups that operated under "ideal circumstances" (striped bars)—that is, when groups followed the provided distribution of instructions instead of manually centralizing, linearizing, or re-distributing among themselves. The "ideal" subset excludes groups whose behavior undermined the experimental manipulation, representing the system's intended use case of collaboration and parallelized work. For each metric, we conducted paired t-tests between conditions and report effect sizes using Cohen's d. Across all 12 groups, all results consistently favored distributed instructions; however, only average first assembly time reached statistical significance ($p = .0126$, $d = 1.28$). In the ideal subset, statistical significance was achieved for both average first assembly time ($p = .0234$, $d = 1.53$) and adjusted completion time ($p = .0304$, $d = 1.35$), reflecting a 77.29% reduction in initial coordination time and 48.61% reduction in error-penalized time. However, error count remained inconclusive ($p = .2570$, $d = .57$). These suggest the system accelerates initial coordination in all cases and overall task completion when the condition is ideal.

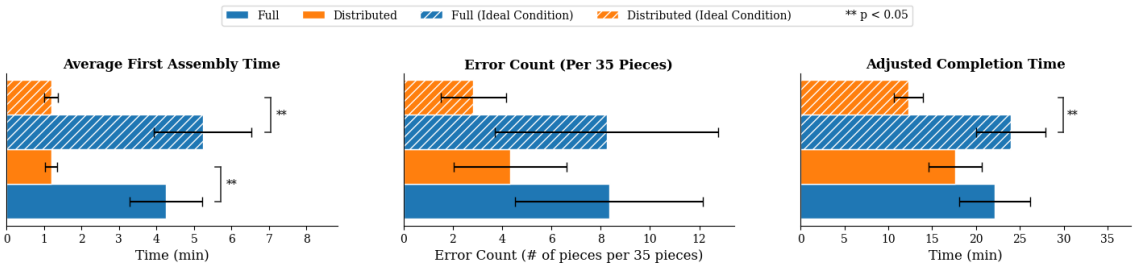

Fig. 1. Performance comparison between full and distributed instruction conditions. Bars show mean values with standard error of the mean. Striped bars indicate performance under ideal conditions. ** p < 0.05.

Perceived Workloads. After completing each task, all members were instructed to complete a post-task survey (Appendix C). Participants rated each dimension (mental demand, physical demand, temporal demand, performance, effort, and frustration) on a scale from 1 to 5, corresponding to the dimensions proposed in NASA-TLX [6]. The results are shown in Figure 2, which displays the average workload scores across all six dimensions for the full and distributed conditions, comparing non-ideal (solid bars) and ideal (striped bars) cases. In all 12 groups, we observed a decrease in perceived workload scores, but paired t-tests revealed no statistically significant differences between the full and distributed conditions across any dimension (all $p > .10$). However, when analyzing only the ideal condition, significant differences emerged in three dimensions: temporal demand ($p = .0012$, $d = 1.85$), performance ($p = .0246$, $d = 1.01$), and effort ($p = .0492$, $d = .84$). In each of these dimensions, the full condition resulted in higher workload scores compared to the distributed condition, suggesting that our distributed instructions reduced overall cognitive burden.

Thematic Analysis. We conducted a reflexive thematic analysis on researcher notes taken on participants' actions and language during tasks and semi-structured interviews, and found that while participants appreciated the efficiency of distributed instructions, they had mixed preferences for instruction type.

LLM-mediated distributed instructions are more efficient than complete instructions. Many groups reported that distributed instructions felt more efficient than full instructions because they enabled parallel work (G6, G7) and allowed members who finished early to assist others (G5, G9).

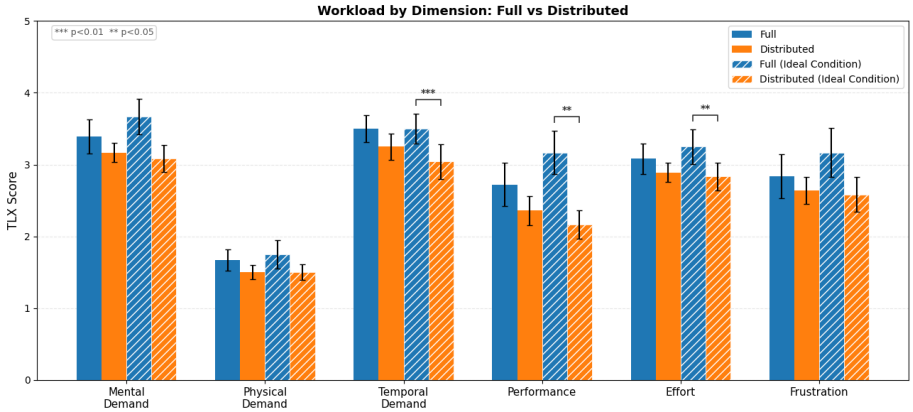

Fig. 2. NASA-TLX score comparison between full and distributed instruction conditions. Bars show mean values with standard error of the mean. Striped bars indicate performance under ideal conditions. *** $p < 0.01$, ** $p < 0.05$.

Long, dense, complete instructions can incur cognitive overload. Several groups reported that using the complete instruction set made it harder to maintain a sense of progress through the task. G2 and G4 noted that they often lost track of their progress due to the long and complex instructions, so they had to review the instructions from the beginning.

Complete instructions create shared understanding, supporting verification and personalized task distribution. Multiple teams noted that complete instruction sets facilitated verification and debugging by providing all members with a shared understanding. G10 described distributed instructions as harder to verify. G12 valued the ability to self-organize roles when using complete instructions, and described assigning roles aligned with individual strengths, such as validator, builder, and material gatherer, after jointly reviewing the complete instruction set.

Distributed instructions complicate error detection and recovery, increasing downstream propagation. While early errors occurred in both conditions, recovery was more difficult under distribution condition as errors introduced by one member were not detectable by others. G1 noted that mistakes had to be located across multiple instruction sets. G10 described an early mistake that led to members misleading each other, guessing, and ultimately proceeding out of order. G12 failed to complete the task after one member's early error went unnoticed.

4.2 Technical Evaluation

To assess generalization and consistency, we conducted an offline evaluation on two representative tasks: the *Red Jet* LEGO build [2] and a multi-step cooking task [14]. For each task, we generated three independent distributed instruction sets using CogniSplit and evaluated them in two stages–required materials and steps–for completeness, accuracy, and specificity.

Stage 1: Are all task materials present and specific? We assessed whether each distributed instruction set explicitly referenced all necessary task materials (i.e., LEGO pieces or cooking ingredients) with sufficient specificity to be actionable. For the LEGO task, all three distributed instruction sets mentioned every required piece. For the cooking task, all sets referenced all required ingredients; however, specificity varied. Instruction set 1 specified quantities for only 12 out of 25 ingredients, whereas instruction sets 2 and 3 specified quantities for 24 and 25 ingredients, respectively.

Stage 2: Are all procedural steps present, accurate, and specific? We evaluated whether each ground-truth procedural step was present in the distributed instructions with accurate description and with sufficient detail. All ground-truth

steps were present and accurate in all distributed instruction sets for both tasks, meaning each set would allow successful task completion. However, specificity varied. For the LEGO task, instruction set 1 contained only 8 out of 14 *highly specific* steps, with the remaining 6 rated *moderately specific*. Set 2 contained 13 out of 14 *highly specific* steps, while set 3 contained all 14. The cooking task was more consistent: all three sets contained 16 out of 16 *highly specific* steps.

5 Discussion

Our results indicate that LLMs can operationalize distributed cognition without requiring teams to explicitly negotiate roles. This finding extends the work of Payne et al. [15], who argue that users offload cognition when the "interaction cost" is low. By using an LLM-mediated system to pre-process and distribute instructions, CogniSplit lowers interaction costs, effectively automating the coordination phase that typically bottlenecks teams. Our work demonstrates that LLMs can function as effective orchestrators, expanding on the work from Jang et al. [9], who identified that "spatially distributed" views reduce the extraneous cognitive load associated with navigating split information. CogniSplit applies this principle to another dimension. By partitioning the instruction set across different users, the system minimizes the extraneous load of filtering out irrelevant tasks, allowing members to focus entirely on their assigned components.

In the subset of teams that adhered to the intended distributed workflow (the "ideal" condition), we observed significant performance gains. By separating the task components, these teams successfully minimized visual interference and capitalized on parallelization, transitioning the workflow from a serial to a parallel process. However, the lack of statistical significance in the full sample's adjusted completion time, temporal demand, performance, and effort attests to the risks of information fragmentation. As identified by Deiglmayr and Schalk [3], fragmenting the core idea of a task can prevent teammates from maintaining a shared mental model. This suggests that while instruction distribution reduces processing load, it must be balanced against the teams' need for shared context to function effectively. Nevertheless, the significant reductions in perceived temporal demand, performance, and effort in the ideal-condition subset align with the task incentive structure (speed and accuracy-based prizes), suggesting CogniSplit's benefits are salient under time pressure. These reductions align with our expectation of reducing cognitive load and expand on prior work in distributed cognition as referred in Sweller and Hutchins' work in human cognition [8, 18].

The significance of CogniSplit lies in bridging the gap between the theoretical benefits of distributed cognition and practical implementation. While prior work demonstrated the utility of distributed interfaces [9, 19] and other work demonstrated the capability of LLMs to coordinate agents [1, 13], our study provides evidence combining these domains for human-human collaboration. We demonstrate that LLMs could serve as orchestrators for the distribution of information, but also highlight that the efficiency of such systems relies heavily on user trust in the fragmented view.

6 Limitations and Future Work

Our findings should be interpreted in light of several limitations. First, LLM output quality varied across generations, with one instruction set under-performing in specificity (Section 4.2). Fine-tuning or reinforcement learning from human feedback could improve consistency. Second, the system provides static instructions with no mechanism for real-time clarification; an interactive mediator allowing users to query step details may reduce confusion. Third, while teams valued self-organizing around individual strengths, CogniSplit assigns roles without accounting for user expertise. Future work should explore adaptive distribution that incorporates user attributes, shifting toward flexible, on-demand support. Finally, our study was limited to LEGO assembly with three-person teams; generalization to other domains and team sizes requires further exploration.

7 Conclusions

While collaborative tasks impose substantial coordination costs, we demonstrate that LLM-mediated instruction distribution can reduce cognitive burden and support collaboration by shifting task decomposition from teams to systems. However, such systems must contend with issues in specificity, error recovery, and shared understanding. CogniSplit serves as a proof-of-concept illustrating both the promise and challenges of using LLMs to distribute collaborative work, pointing toward future opportunities for personalization and diverse coordination styles.

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